# Design of a Real-Time, IoT-Based Dog Feeding System with ESP32 and Voice Assistance

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***Abstract*— The Real-Time IoT-Based Dog Feeding System ensures timely and consistent feeding of pets by automating the process using ESP32 and voice assistance technology. This system eliminates the need for cloud connectivity by operating completely offline, thus making it suitable for both urban and rural environments. It integrates components such as a servo motor for mechanical food dispensing, DFPlayer Mini with a speaker for audio guidance, and 16 × 2 LCD display for real-time status updates. The ESP32 microcontroller acted as the central unit, coordinating the components to dispense food at set intervals. When activated, the system rotates the servo to release a predefined amount of food into the bowl while simultaneously playing a voice message to indicate successful operation. The LCD displays system messages, such as feeding status or alerts. By offering scheduled and consistent feeding without requiring Internet or mobile app connectivity, the system supports pet owners who may be away or have irregular schedules. User-friendly design and voice feedback enhance accessibility, especially for children and visually impaired individuals. Overall, the proposed system provided an efficient, low-cost, and reliable solution for automated pet care.**

**Keywords— ESP32, IoT, Dog Feeder, DFPlayer Mini, Voice Assistant, Servo Motor, LCD Display, Real-Time System**

## *I. Introduction*

Pet care has become an essential concern in modern households, especially for working professionals who are unable to consistently manage feeding schedules. Inconsistent feeding can lead to health issues such as obesity, malnutrition, and digestive problems in pets. To address this challenge, automated feeding systems have emerged as an effective solution to maintain a consistent feeding routine, ensuring pets receive the right quantity of food at the right time without human intervention. The proposed system in this paper is a **real-time, IoT-based dog feeding system** using the **ESP32 microcontroller**, designed to be efficient, affordable, and user-friendly.

While many existing smart pet feeders rely on cloud-based platforms and mobile apps for operation, they often suffer from internet connectivity issues and higher costs. In contrast, the presented system is **completely offline**, powered via a stable DC power supply, and does not depend on cloud storage or wireless communication. The system uses **DFPlayer Mini and a speaker** for **voice-based assistance**, making interactions intuitive. It offers audio cues to notify when food is being dispensed, enhancing the interactivity and responsiveness of the system.

The feeding mechanism is managed using a **servo motor**, which rotates to dispense a predefined quantity of food. The status of the operation—such as "Feeding Started" or "Bowl is Full"—is displayed on a **16x2 LCD**, which serves as a real-time feedback interface. This enables users to quickly identify the system state. The core controller, the **ESP32**, plays a crucial role in synchronizing the motor, speaker, and display module to execute feeding at regular time intervals or on user command.

Voice feedback enhances accessibility, especially for visually impaired pet owners or children. Additionally, since the system avoids internet dependency, it ensures robust and **reliable operation during power or network outages**, making it suitable for rural or remote environments. The system prioritizes safety and longevity by supplying stable voltage levels using a DC power converter to prevent damage to sensitive modules.

This project aims to provide a cost-effective, efficient, and fully autonomous pet feeding solution without the need for advanced user knowledge or interaction. By automating the feeding process, the system reduces stress for pet owners and ensures pets are consistently and healthily fed. The novelty of the system lies in its **offline automation, real-time voice interaction**, and **modular design**, making it adaptable and extensible for future improvements such as weight sensing or scheduling via RTC modules. This paper discusses the system architecture, implementation, and performance analysis of the proposed pet feeder.

## *II. RELATED WORKS*

Various methodologies have been implemented to automate pet feeding systems using IoT components, machine learning techniques, and sensor integrations. The following section reviews related systems and their methodologies, covering automation mechanisms, hardware components used, and feedback features.

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An IoT-based pet feeder utilizing the ESP32 microcontroller and deep learning techniques was developed to dispense appropriate food portions, enhancing pet nutrition management and overall health. The system integrates sensors such as the HX711 weight sensor and HC-SR04 ultrasonic sensor, along with a camera module for pet identification. The ESP32 manages the coordination of these components, ensuring accurate and timely feeding. The web-based application, developed using PHP and HTML, allows users to control the feeder remotely and monitor real-time data linked with a Firebase database. This approach addresses challenges like overfeeding and underfeeding, contributing to improved pet well-being.

Another approach involves transforming an ESP32 into a smart pet feeder, enabling remote and scheduled feeding through IoT technology. The system employs a stepper motor (NEMA 17) controlled by an L298N motor driver to dispense food accurately. Users can schedule feeding times, monitor food levels, and control the feeder via a mobile app or web interface. Optional sensors, such as weight sensors or cameras, can be integrated for enhanced monitoring. This project demonstrates the feasibility of creating a functional and reliable pet feeder using readily available components.

An IoT pet feeder was developed using the ESP8266 module and the Blynk mobile app, allowing pet owners to feed their pets remotely. The system features a stepper motor to control the feeding mechanism, with the Blynk app providing an intuitive interface for scheduling and manual feeding. The app displays real-time data, including feeding times and hopper levels, ensuring pet owners are informed about their pet's feeding status. This implementation highlights the practicality of integrating mobile applications with IoT devices for pet care.

A smart pet feeder project utilizing ESP32 and STM8 nodes was designed to enable remote-controlled feeding and entertainment of pets via a web interface. The system comprises a master node (ESP32) that manages communication with slave nodes (STM8), which handle feeding mechanics, motion detection, and audio playback. Wireless communication is facilitated using the nRF24 module, ensuring robust and low-power data exchange between nodes. This architecture allows for scalable and efficient pet feeding solutions.

An IoT-based pet feeder using the NodeMCU-32S microcontroller was implemented to allow remote feeding of pets. The system employs an infrared sensor to detect the pet's presence near the food container, notifying the owner via a dashboard. A servo motor controls the opening of the food container, which can be operated remotely through the V-One dashboard. This setup provides a cost-effective solution for pet owners to manage feeding schedules conveniently.

A Google Assistant-controlled IoT-based pet feeder was developed using the NodeMCU microcontroller, enabling remote feeding via voice commands. Users can issue commands like "Okay Google, feed my pet" to dispense food, offering convenience and care for pets even when owners are away. This integration of voice assistants with IoT devices showcases the potential for seamless and user-friendly pet care solutions.

A smart pet feeder project employed an ESP8266 microcontroller and Blynk mobile application to create an automated feeding system. The feeder can dispense food at scheduled times or manually via the Blynk application, providing flexibility for pet owners. The system includes a stepper motor to control the dispensing mechanism and utilizes the ESP8266 for wireless communication. This setup demonstrates the effectiveness of combining IoT technology with mobile applications for pet care.

Another implementation involved developing a smart pet feeder using an ESP8266 microcontroller, integrating features such as presence detection and scheduled feeding. The system uses a PIR sensor to detect the pet's presence and a beeper to notify the pet about mealtimes. Activation can occur locally or via Wi-Fi, allowing for remote control through a web interface. This design emphasizes the importance of user-friendly interfaces and real-time monitoring in automated pet feeding systems.

An innovative approach utilized the ESP8266 microcontroller and MQTT protocol to create an automated pet feeder. The system allows the scheduling of feeding times and manual activation via a mobile application, ensuring consistent feeding routines. The integration of MQTT enables efficient communication between the feeder and the mobile app, highlighting the potential of IoT protocols in pet care automation.

These studies demonstrate the effectiveness of integrating IoT technologies, such as ESP32 and ESP8266 microcontrollers, with various sensors and actuators to create automated pet feeding systems. By leveraging components like weight sensors, ultrasonic sensors, cameras, and servo motors, these systems ensure accurate and timely feeding. The incorporation of web-based applications and mobile interfaces allows pet owners to monitor and control feeding schedules remotely, enhancing convenience and promoting the health and well-being of pets.

## *III. METHODOLOGY*

A real-time dog feeding system was built using the ESP32 microcontroller as the central unit. The ESP32 was connected to a servo motor that controlled the opening and closing of the food container. A 16x2 LCD is used to display system messages such as “Feeding Started” or “Feeding Completed.” To provide voice feedback, a DFPlayer Mini module was connected to a speaker, which played pre-recorded audio files during feeding. The system is powered by a 5V DC power adapter, ensuring continuous and stable operation without relying on batteries or internet connectivity.

When a feeding command is given, either manually via a button or based on a preset schedule, the ESP32 triggers the servo motor to rotate, dispensing a fixed amount of food. Simultaneously, it sends commands to the DFPlayer Mini to play voice messages and update the LCD with the current status. All components are integrated into a single circuit, and the system runs offline, making it highly suitable for home use, particularly in rural or low-network areas.

## *IV. SYSTEM DESIGN*

The IoT-based dog feeding system is designed using a combination of embedded hardware modules and mobile-app-based control to automate pet feeding. The system consists of five key hardware units and a microcontroller programmed to execute feeding commands, trigger voice feedback, and display system status. It is powered via a direct 5V DC cable, enabling consistent operation without relying on batteries or cloud services.

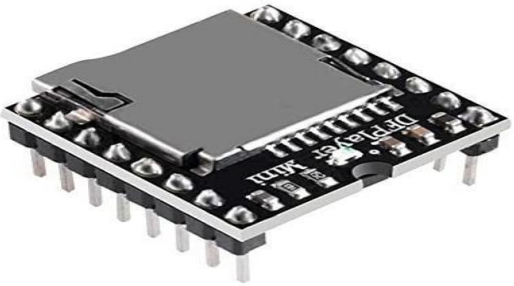
*Hardware Components:*

ESP32 Microcontroller: The ESP32 served as the central control unit of the system. It processes input signals from the mobile app (via virtual buttons on Blynk) and manually connected hardware

buttons and executes programmed logic to dispense food. ESP32 also controls the voice output and display units, ensuring that all components work in coordination. Its GPIO pins were used to drive the servo motor, communicate with the DFPlayer Mini, and update the LCD in real time.

ServoMotor: A standard 180-degree servo motor was used to rotate the food dispenser door. It was programmed to turn to a specific angle for a short duration (e.g., 90° rotation), releasing a fixed portion of food into the pet's bowl. Once the rotation was complete, the servo reset to its original position to prevent continuous food flow. The ESP32 controls the motor using precise PWM signals.

Speaker (DFPlayer Mini): The DFPlayer Mini module is connected to a speaker to play pre-recorded audio messages
such as "Feeding started" or "Food is ready." These messages are stored on a microSD card inside the module and played by sending serial commands from the ESP32. This feature helps alert the pet when food is dispensed and makes the system interactive and user-friendly



LCD Display (16x2): The 16x2 LCD is used to display real-time system updates like "Waiting," "Feeding Now," or "Feeding Done." This helps users visually confirm the operational state of the feeder. The LCD is connected to the ESP32 using either the 4-bit mode or the I2C, depending on the version, and updates automatically with each cycle.

Power Supply: The entire system is powered using a 5V regulated DC adapter. This ensures that the system can operate continuously without interruption, unlike battery-operated models. The adapter is chosen to match the current requirements of all connected modules.

*Software Components:*

Blynk Mobile App Interface: The system uses the Blynk platform to create a user-friendly mobile interface for controlling the dog feeder. Users can press a virtual button on the app to initiate the feeding process remotely. The ESP32 is connected to the Blynk app via Wi-Fi and listens for input on a virtual pin assigned to the button. This feature allows for flexible, wireless operation without physical interaction.

Embedded Firmware: The ESP32 was programmed using an Arduino IDE. The code handles all interactions, including reading the Blynk virtual button, controlling the servo motor, sending UART commands to the DFPlayer Mini for audio playback, and updating the LCD. Debouncing, safety delays, and timing logic were implemented in the code to ensure correct dispensing behaviour and prevent multiple triggers within a short span.

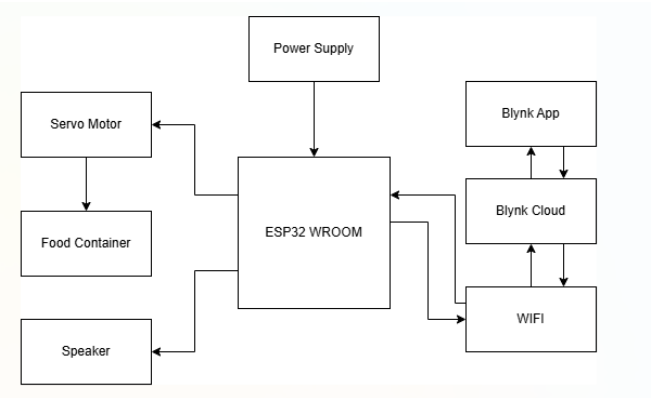
Voice Assistance Logic: The software triggers predefined audio tracks stored on the DFPlayer Mini when the feeding is in progress. The tracks are assigned to numbered files on the SD card, and the ESP32 sends play commands via serial communication. This makes the system more engaging with pets and provides audio confirmation to users.

System Timing and Safety Logic: Although the system does not use an RTC or cloud sync, internal timing logic is used in the ESP32 code to avoid double feeds. Delays and reset flags are maintained in memory to ensure the servo cannot be retriggered within a fixed duration, protecting against accidental multiple feed cycles.

## *V. BLOCK DIAGRAM*

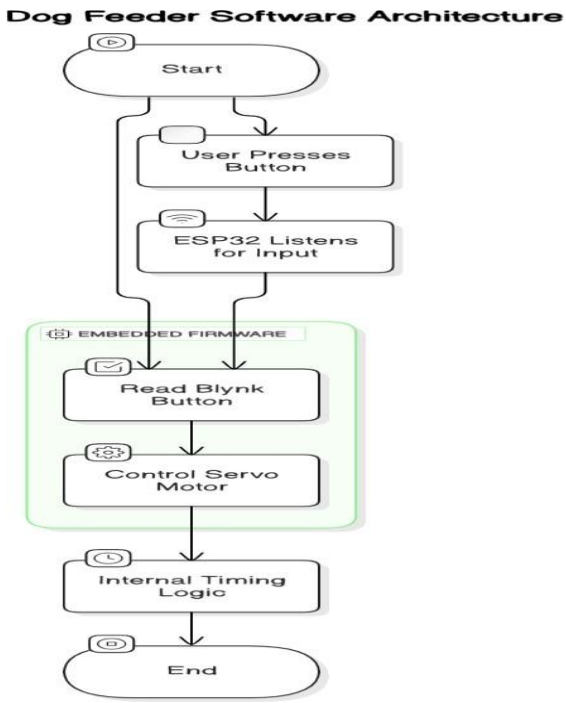
### *A. Hardware Design*

The hardware block diagram shows how the ESP32 WROOM microcontroller controls the dog-feeding system by operating a servo motor, speaker, and LCD. It connects to Wi-Fi and the Blynk App, allowing users to remotely control and monitor feeding.




### *B. Software Design*

The software was developed using an Arduino IDE to program the ESP32, which connects to the Blynk app via Wi-Fi. It controls the servo motor, plays voice messages, updates the LCD, and uses timing logic to ensure safe single-time feeding.




## *VI . CONCLUSION*

The real-time IoT-based dog feeding system using ESP32 and voice assistance provides an efficient offline solution for automating pet care. By integrating a servo motor, LCD display, and DFPlayer Mini, the system offers mechanical food-dispensing, real-time feedback, and voice alerts that enhance the user and pet experience. Operating on a direct power supply, the system is designed for reliability, especially in regions with limited internet connectivity. The inclusion of a mobile interface via the Blynk app allows convenient remote control, making it suitable for busy pet owners. The system ensures timely and measured feeding, supports interactions through audio feedback, and is easy to maintain and deploy.

Overall, the project achieved its objective of delivering a low-cost, interactive, and autonomous dog feeding system. Its simplicity and modular design enable further upgrading and ensure adaptability to various user needs. This solution demonstrates how embedded IoT systems can be applied in real-life scenarios to improve pet care and automation at home.

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## *VII.FUTURE ENHANCEMENTS*

In future iterations, the system can be extended with additional features to improve hygiene and functionality. One of the proposed enhancements is the **automatic cleaning of the pet bowl** after each feeding cycle. This can be implemented using a small water pump, nozzle, and drainage system that is activated after food is dispensed. Cleaning can be automated through a relay-controlled water pump triggered by the ESP32, ensuring that the bowl remains sanitary between meals.

Additionally, a **real-time clock (RTC)** module can be added to support accurate scheduled feeding even in offline mode, eliminating the need for mobile-triggered actions. A **weight sensor** may also be integrated to measure the amount of food dispensed and to monitor consumption patterns. These enhancements will make the system more intelligent, hygienic, and self-sufficient, thereby moving closer to a fully automated pet care solution.

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## REFERENCES

[1] M. Rohs and B. Gfeller, “Using Camera-Equipped Mo-bile Phones for Interacting with Real-World Object,” Proceedings of Advances in Pervasive Computing, April 2004, pp. 265-271.

[2] C. Sammarco and A. Lera, “Improving Service Manage-ment in the Internet of Things,” Sensors, Vol. 12, No. 9, 2012, pp. 11888-11909. doi:10.3390/s120911888

[3] H. Ning and H. Liu, “Cyber-Physicl-Social Based Secu-rity Architecture for Future Internet of Things,” Advanced in Internet of Things, Vol. 2, No. 1, 2012, pp. 1-7. doi:10.4236/ait.2012.21001

[4] M. Kranz, P. Holleis and A. Schmidt, “Embedded Inter-action Interacting with the Internet of Things,” IEEE Internet Computing, Vol. 14, No. 2, 2010, pp. 46-53. doi:10.1109/MIC.2009.141

[5] G. Kortuen, F. Kawsar, D. Fitton and V. Sundramoorthy, “Smart Objects as Building Blocks for the Internet of Things,” IEEE Internet Computing, Vol. 14, No. 1, 2010, pp. 44-51. doi:10.1109/MIC.2009.143

[6] I. F. Akyildiz, W. Su, Y. Sankar Subramaniam and E. Cayirci, “Wireless Sensor Networks: A Survey,” Com- puter Networks, Vol. 38, No. 4, 2002, pp. 393-422. doi:10.1016/S1389-1286(01)00302-4

[7] Vania, Kanisius Karyono, Hargyo Tri Nugroho. Computer Engineering, Multimedia Nusantara University, Tangerang, Indonesia: Smart Dog Feeder Design Using Wireless Communication, MQTT and Android Client 2016 International Conference on Computer,Control,Informatics and its Applications,DOI: 10.1109/IC3INA.2016.7863048, 978-1-5090-2323-3/16/$31.00 c 2016 IEEE.

[8] Tessema Gelila Berhan, Worku Toyiba Ahemed, Tessema Zelalem Birhan. Tianjin University of Technology and Education (TUTE), School of Electronics Engineering, Tianjin: Programmable Pet Feeder, International Journal of Scientific Engineering and Research (IJSER), Volume 3 Issue 11, November 2015 ISSN (Online): 2347-3878, Impact Factor (2014): 3.05.

[9] Vania, "Perancangan Smart Dog Feeder Menggunakan Protokol Komunikasi WiFidan MQTT Serta Client Berbasis Android," Universitas Multimedia Nusantara, Tangerang, Skripsi 2016.

[10] Even Yehudah Ofer Landau, "Dry Food Dispensing System," US 6,964,355 B2, Nov. 15, 2005.

[11] Luigi Atzori, Antonio Iera, Giacomo Morabito, "The Internet of Things: A Survey", Elsevier, May 2010

[12] Saurabh A. Yadav, Sneha S. Kulkarni, Ashwini S. Jadhav, Prof. Akshay R. Jain, Computer engineering, PVGCOEN, Maharashtra, INDIA, "IOT BASED PET FEEDER SYSTEM", Vol-4 Issue-2 2018 IJARIIE-ISSN(O)-2395-4396.

[13] Industrial IoT Adoption, Deloitte Insights 2023: www2.deloitte.com/iot-industrial

[14] Yixing Chen, Maher Elshakankiri Department of Computer Science University of Regina Regina, Canada, "Implementation of an IoT based Pet Care System", 2020 Fifth International Conference on Fog and Mobile Edge

[15] A. Luayon, G. Tolentino, V. Almazan, P. Pascual, and M. Samonte, "PetCare: A Smart Pet Care IoT Mobile Application", in IC4E 2019: 2019 10th International Conference on EEducation, E-Business, management and E-Learning, Tokyo, 2019, pp. 427–431

[16] S.Subaashri, M.Sowndarya, D.K.S. Sowmiyalaxmi, S.V.Sivassan, and C. Rajasekaran, "Automatic Pet Monitoring and Feeding System Using IoT", International Journal of ChemTech Research, vol. 10, no. 14, pp. 253-258, 2017.

[17] Priya Mondal, Dr. Swapnili Karmore, Rajnandnee Parnami Mukt Shabd Journal, "Design and development of IoT based Smart Pet Feeder", ISSN NO: 2347-3150, Volume IX, Issue V, MAY/2020.

[18] Vrishanka, P.N.; Prabhakar, P.; Shet, D.; Rupali, K. Automated Pet Feeder using IoT. In Proceedings of the 2021 IEEE International Conference on Mobile Networks and Wireless Communications (ICMNWC), Tumkur, Karnataka, India, 3–4 December 2021; pp. 1–5.

[19] Mritunjay Subhashchandra Tiwari, Sahil Manoj Hawal, Nikhil Navanath Mhatre, Akshay Ramesh Bhonsale, Mainak Bhaumik, "Automatic Pet Feeder Using Arduino, International Journal of Innovative Research in Science, Engineering and Technology", vol. 7, no. 3, March 2018.